UNIVERSIDADE DE SÃO PAULO
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Thermal Relaxation Asymmetry in Quantum Systems

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Primeira frase do agradecimento ....

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*"O estudo, a busca da verdade e da beleza são domínios
em que nos é consentido sermos crianças por toda a vida."*
Albert Einstein

ABSTRACT

RAUPP, C. **Thermal Relaxation Asymmetry in Quantum Systems**. 2026. 79 p. Dissertation (Master in Science) - Instituto de Física de São Carlos, Universidade de São Paulo, São Carlos, 2026.

This is the english abstract.

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RESUMO

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LIST OF ABBREVIATIONS AND ACRONYMS

ABNT	Associação Brasileira de Normas Técnicas
abnTeX	ABsurdas Normas para TeX
IBGE	Instituto Brasileiro de Geografia e Estatística
LaTeX	Lamport TeX
USP	Universidade de São Paulo
USPSC	Campus USP de São Carlos

LIST OF SYMBOLS

Γ	Letra grega Gama
Λ	Lambda
ζ	Letra grega minúscula zeta
$\in$	Pertence

CONTENTS

1 INTRODUÇÃO

2 INTRODUCTION TO OPEN QUANTUM SYSTEMS THEORY

This chapter introduces how quantum systems evolve in time. We begin discussing the unitary evolution of closed quantum systems in Sec. 2.1. However, this description cannot be applied to open quantum systems. A solution for this is presented in Sec. 2.2, where we present a Markovian master equation and specify its form to describe a thermalisation process. The content of this chapter are based on Refs. (1–3).

2.1 Dynamics of Close Quantum Systems

Quantum mechanics was initially formulated for isolated systems, whose physical states are represented as a normalised vector $|\psi\rangle$ in the Hilbert space $\mathcal{H}$. This representation contains all the information about the physical state, called a pure state. However, another mathematically equivalent formulation is more convenient for describing quantum systems whose states are not fully known. In these cases, it is only possible to assign a probability p_i of the system existing in the state $|\psi_i\rangle$, which is encapsulated in the density operator,

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right| . \tag{2.1}$$

The pure state is a special case in the density operator notation, where it is written as $\hat{\rho} = |\psi\rangle \langle\psi|$. The density operator (also called density matrix) must satisfy the following conditions: (1) the sum of all p_i is equal to one, which is equivalent to $\mathrm{Tr}[\hat{\rho}] = 1$, and (2) it is positive semi-definite, which means all $\hat{\rho}$ eigenvalues are $\lambda_k \geq 0$.

More than describing physical states, quantum mechanics explains how quantum systems evolve in time. For closed quantum systems, the time evolution is given by the Schrödinger equation,

$$i\hbar \frac{d}{dt} \left|\psi(t)\right\rangle = \hat{H}(t) \left|\psi(t)\right\rangle , \tag{2.2}$$

where $\hat{H}(t)$ is the system's Hamiltonian and Planck's constant will be $\hbar = 1$ from now on. This equation can be translated into the density matrices notation as the Liouville-von Neumann equation,

$$\frac{d}{dt}\hat{\rho}(t) = -i[\hat{H}(t), \hat{\rho}(t)]. \tag{2.3}$$

Its solution describes the unitary evolution of an initial state $\hat{\rho}(0)$ as

$$\hat{\rho}(t) = U(t)\hat{\rho}(0)U^{\dagger}(t), \tag{2.4}$$

where $U(t)$ is a unitary operator with properties $U^{-1} = U^{\dagger}$ and $UU^{\dagger} = \mathbb{I}$, in which $\mathbb{I}$ is the identity matrix. Unitary dynamics conserve the total probability, which makes sense as an isolated system cannot lose any information. A consequence of preserving the information is that the dynamics are time-reversible.

Even though all this formalism describes a variety of quantum phenomena, it is an idealisation because every physical system interacts with others. So, let's take a step forward and consider a quantum system (S) interacting with another system called the environment (E). Together, they form a closed quantum system that evolves in time by the unitary dynamics described before. The total system ($S+E$) is defined by the Hamiltonian

$$\hat{H}_{SE} = \hat{H}_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes \hat{H}_E + \hat{H}_I, \tag{2.5}$$

where $\hat{H}_S$ is the system's Hamiltonian, $\hat{H}_E$ is the environment's Hamiltonian, and $\hat{H}_I$ describes the system-environment interaction. Some characteristics of the environment are that it does not evolve with time and is usually considered to have infinite degrees of freedom, over which we cannot access information or have control. As a result, the system S is the only part we can manipulate, being the focus of our studies. However, because the system S exchanges information with the environment, it does not have a unitary dynamics in general. This is what we call an open quantum system, illustrated in Fig. 1, whose dynamics will be introduced in the next section.

2.2 Dynamics of Open Quantum Systems

In this section, we discuss the fundamental steps to obtain the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation, also called the Lindblad master equation for short. Let's begin with the time evolution of the total system ($S+E$). We know it is unitary, meaning that the Liouville-von Neumann equation can be used. However, we do not have access to all the information in the environment, and including its dynamics in the calculation can be overwhelming. Fortunately, it is possible to make some approximations

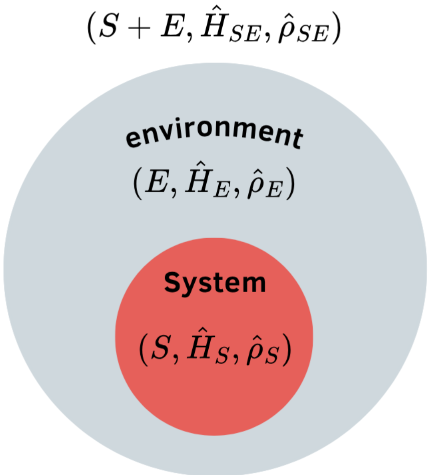

Figure 1 – Scheme of an open quantum system.

to describe the system's dynamics, taking into account environmental effects, without needing to deal with the environment's time evolution. These approximations become a bit easier in the interaction picture, where both states and operators change with time. Up to this point, we have used the Schrödinger picture, in which the state evolves with time while the operators remain unchanged. To convert from the Schrödinger picture to the interaction picture, we split the total Hamiltonian as $H_{SE} = H_0 + H_I$ with $H_0 = \hat{H}_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes \hat{H}_E$, and the time evolution of an operator $\hat{O}$ becomes $\hat{O}(t) = e^{iH_0 t}\hat{O}e^{-iH_0 t}$. As a result, the Liouville-von Neumann equation describing the total system-environment dynamics becomes

$$\frac{d}{dt}\hat{\rho}_{SE}(t) = -i[\hat{H}_I(t), \hat{\rho}_{SE}(t)]. \tag{2.6}$$

We can integrate this equation and substitute into itself, obtaining

$$\frac{d}{dt}\hat{\rho}_{SE}(t) = -i[\hat{H}_I(t), \hat{\rho}_{SE}(0)] - \int_0^t ds[\hat{H}_I(t), [\hat{H}_I(s), \hat{\rho}_{SE}(s)]]. \tag{2.7}$$

Because we are only interested in the system's dynamics, we can erase the environment by taking the partial trace. All this algebra still has not solved the problem with environmental dynamics, so our next steps are making approximations. First, we will consider a weak system-environment interaction, allowing us to use the Born approximation, $\hat{\rho}_{SE}(t) \approx \hat{\rho}_S(t) \otimes \hat{\rho}_E$. An interesting consequence of using the weak coupling is that the environment is essentially unaffected by the system S, so it can be considered memoryless, allowing us to apply the Markov approximation. In other words, the time the environment takes to erase the system's perturbation is much shorter than the relaxation time, which is equivalent to extending $t \to \infty$. By using the Markovian approximation, the system's time evolution does not depend on its past states, which means $\rho_S(s) = \rho_S(t)$. After all these approximations, we find the Redfield equation,

$$\frac{d}{dt}\hat{\rho}_S(t) = -i\mathrm{Tr}_E[\hat{H}_I(t), \hat{\rho}_S(0) \otimes \hat{\rho}_E] - \mathrm{Tr}_E \int_0^\infty ds[\hat{H}_I(t), [\hat{H}_I(s), \hat{\rho}_S(t) \otimes \hat{\rho}_E]]. \tag{2.8}$$

In most cases, the mean of environmental fluctuations is zero, which makes the first term zero. So, we will focus our analysis on the second term.

Although Eq. (2.8) can describe the relaxation of open quantum systems, it does not guarantee the positivity of probabilities, which can lead to unphysical results. Then, a solution is to use one last approximation: the rotating wave approximation (RWA). In this section, we will use a simpler scenario to make the RWA, but the general case can be found in (1). Let's consider that a perturbation of the bath results in a single effect on the system, that is, no multiple transitions can occur simultaneously in the system. So, the interaction Hamiltonian in the Schrödinger picture can be decomposed into two Hermitian operators as

$$\hat{H}_I = \hat{A} \otimes \hat{B}, \tag{2.9}$$

The next step is to write $\hat{A}$ in terms of the eigenstates of the system's Hamiltonian, $\hat{H}_S |n\rangle = \lambda_n |n\rangle$. To make the RWA, we will separate this operator as $\hat{A} = \sum_\omega \hat{A}(\omega)$, where

$$\hat{A}(\omega) = \sum_{\substack{n,m \\ \omega=\lambda_m-\lambda_n}} |n\rangle \langle n| \hat{A} |m\rangle \langle m| \tag{2.10}$$

is summing over all combinations of states that have frequency ω. In the interaction picture, this Hamiltonian becomes

$$\hat{H}_I(t) = \sum_\omega e^{-i\omega t} \hat{A}(\omega) \otimes \hat{B}(t) = \sum_{\omega'} e^{i\omega' t} \hat{A}^\dagger(\omega') \otimes \hat{B}^\dagger(t). \tag{2.11}$$

The last equality came from using the property $A(\omega) = A^\dagger(-\omega)$ and $\omega' = -\omega$. With this decomposition of the interaction Hamiltonian, we use $s \to t - s$ and define

$$\hat{H}_I(t-s) = \sum_\omega e^{-i\omega(t-s)} \hat{A}(\omega) \otimes \hat{B}(t-s), \tag{2.12}$$

$$\hat{H}_I(t) = \sum_{\omega'} e^{i\omega' t} \hat{A}^\dagger(\omega') \otimes \hat{B}^\dagger(t), \tag{2.13}$$

to rewrite Eq. (2.8) as

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_S(t) &= -\mathrm{Tr}_E \int_0^\infty ds [\hat{H}_I(t), [\hat{H}_I(t-s), \hat{\rho}_S(t) \otimes \hat{\rho}_E]] \\ &= \sum_{\omega,\omega'} e^{i(\omega'-\omega)t} \Gamma(\omega)(\hat{A}(\omega)\hat{\rho}_S(t)\hat{A}^\dagger(\omega') - \hat{A}^\dagger(\omega')\hat{A}(\omega)\hat{\rho}_S(t)) + \text{h.c}, \end{aligned} \tag{2.14}$$

where

$$\Gamma(\omega) = \int_0^\infty ds e^{i\omega s} \langle \hat{B}^\dagger(t)\hat{B}(t-s)\rangle_E, \tag{2.15}$$

with $\langle \hat{B}^\dagger(s)\hat{B}(0)\rangle_E = \mathrm{Tr}[\hat{B}^\dagger(s)\hat{B}(0)\hat{\rho}_E]$ being the correlation functions of the bath. The rotating wave approximation imposes that the terms with $\omega \neq \omega'$ oscillate so fast that they do not affect the relaxation process. As a result, only the cases of resonance ($\omega = \omega'$) contribute to the equation. Thus we have

$$\frac{d}{dt}\hat{\rho}_S(t) = \sum_\omega \Gamma(\omega)(\hat{A}(\omega)\hat{\rho}_S(t)\hat{A}^\dagger(\omega) - \hat{A}^\dagger(\omega)\hat{A}(\omega)\hat{\rho}_S(t)) + \text{h.c.} \tag{2.16}$$

One last step to obtain the usual form of the Lindblad master equation is to separate $\Gamma(\omega)$ into its real and imaginary parts as

$$\Gamma(\omega) = \frac{1}{2}\gamma(\omega) + iS(\omega), \tag{2.17}$$

where the transition rate is

$$\gamma(\omega) = \int_{-\infty}^\infty ds e^{i\omega s} \langle \hat{B}^\dagger(s)\hat{B}(0)\rangle_E, \tag{2.18}$$

and $S(\omega)$ produce slight changes on the system's energy levels, which are included in the Lamb shift Hamiltonian

$$\hat{H}_{LS} = \sum_{\omega} S(\omega)\hat{A}^{\dagger}(\omega)\hat{A}(\omega). \tag{2.19}$$

As a result, the Lindblad master equation in the interaction picture has the form

$$\frac{d}{dt}\hat{\rho}_S(t) = -i[\hat{H}_{LS}, \hat{\rho}_S(t)] + \mathcal{D}(\hat{\rho}_S(t)), \tag{2.20}$$

with the dissipator being

$$\mathcal{D}(\hat{\rho}_S) = \sum_{\omega} \gamma(\omega)\left(\hat{A}(\omega)\hat{\rho}_S\hat{A}^{\dagger}(\omega) - \frac{1}{2}\{\hat{A}^{\dagger}(\omega)\hat{A}(\omega), \hat{\rho}_S\}\right), \tag{2.21}$$

where $\hat{A}(\omega)$ are usually called jump operators. In the Schrödinger picture, the Lindblad master equation becomes

$$\frac{d}{dt}\hat{\rho}_S(t) = -i[\hat{H}_S + \hat{H}_{LS}, \hat{\rho}_S(t)] + \mathcal{D}(\hat{\rho}_S(t)), \tag{2.22}$$

where the jump operators remain the same. It is very common to neglect the Lamb shift Hamiltonian, because it only adds a constant to the expression, without changing the physics of the system. That said, from now on, we will use only the $\hat{H}_S$ in our Lindblad master equation.

2.2.1 Davies Maps

An important class of Lindblad master equations is the Davies maps, which generate the thermalisation of a quantum system weakly coupled to a thermal bath at inverse temperature β (4). This relaxation process always results in thermal states, which means the Davies maps' steady state is always a Gibbs state of the form

$$\hat{\rho}_{ss} = e^{-\beta\hat{H}_S}/\mathcal{Z}_S, \tag{2.23}$$

where $\mathcal{Z}_S = \mathrm{Tr}[\exp(-\beta\hat{H}_S)]$ is the partition function. For this to be true, the transition rates $\gamma(\omega)$ must obey the detailed balance condition,

$$\gamma(-\omega) = \gamma(\omega)e^{-\beta\omega}. \tag{2.24}$$

Given these conditions, the jump operators of the Davies map have a specific format. Let's rewrite the dissipator (Eq. (2.21)) as

$$\mathcal{D}(\hat{\rho}_S) = \sum_{\omega} \hat{L}_{\omega}\hat{\rho}_S\hat{L}_{\omega}^{\dagger} - \frac{1}{2}\{\hat{L}_{\omega}^{\dagger}\hat{L}_{\omega}, \hat{\rho}_S\}, \tag{2.25}$$

defining the jump operators as $\hat{L}_{\omega} = \sqrt{\gamma(\omega)}\hat{A}(\omega)$. To compute the transition rate ($\gamma(\omega)$), according to Eq. (2.18), we need to use that our environment is a thermal bath. The thermal bath can be described by a set of infinite independent harmonic oscillators, $\hat{H}_E = \sum_k \omega_k \hat{b}_k^\dagger \hat{b}_k$, where ω_k is the frequency of the k-mode, and $\hat{b}_k$ and $\hat{b}_k^\dagger$ are the annihilation and creation operators of the k-mode. That said, the hermitian operator from Eq. (2.9) $\hat{B} = \sum_k (g_k \hat{b}_k^\dagger + g_k^* \hat{b}_k)$ in the interaction picture is

$$\hat{B}(s) = \sum_k (g_k \hat{b}_k^\dagger e^{i\omega_k s} + g_k^* \hat{b}_k e^{-i\omega_k s}), \tag{2.26}$$

with g_k being the system-environment coupling parameter. The density matrix of the thermal bath is

$$\hat{\rho}_E = \prod_i (1 - e^{-\beta\omega_i}) e^{-\beta\omega_i \hat{b}_i^\dagger \hat{b}_i}, \tag{2.27}$$

with which we can compute the correlation functions, obtaining

$$\langle \hat{B}^\dagger(s)\hat{B}(0)\rangle_E = \sum_k |g_k|^2 (e^{-i\omega_k s}(\bar{n}(\omega_k) + 1) + e^{i\omega_k s}\bar{n}(\omega_k)), \tag{2.28}$$

where $\bar{n}(\omega_k) = (e^{\beta\omega_k} - 1)^{-1}$ is the Bose-Einstein distribution. Then, we find the transition rates by solving the integral in Eq. (2.18), resulting in

$$\gamma(-\omega) = 2\pi \sum_k |g_k|^2 \bar{n}(\omega_k)\delta(\omega - \omega_k) \tag{2.29}$$

$$\gamma(+\omega) = 2\pi \sum_k |g_k|^2 (\bar{n}(\omega_k) + 1)\delta(\omega - \omega_k). \tag{2.30}$$

Assuming that the infinite harmonic oscillators of the bath have close frequencies, we can convert from discrete to continuum space by changing the sum to an integral in ω, g_k to a function $g(\omega)$ and including the density function of the bath's states $d(\omega)$. The solution of this new integral gives transition rates that respect the detailed balance (Eq. (2.24)),

$$\gamma(-\omega) = \Gamma(\omega)\bar{n}(\omega), \tag{2.31}$$

$$\gamma(+\omega) = \Gamma(\omega)(\bar{n}(\omega) + 1), \tag{2.32}$$

where $\Gamma(\omega) = 2\pi d(\omega)|g(\omega)|^2$. Considering a homogeneous bath and the same coupling for all k-modes, Γ becomes a constant. Also, because we have two frequencies, we solve the sum in ω from the Eq. (2.25), obtaining the following Lindblad equation for the Davies maps,

$$\frac{d}{dt}\hat{\rho}_S(t) = -i[\hat{H}_S, \hat{\rho}_S(t)] + \mathcal{D}^-(\hat{\rho}_S) + \mathcal{D}^+(\hat{\rho}_S), \tag{2.33}$$

where each dissipator's jump operator is

$$L_\omega^+ = \sqrt{\Gamma(\bar{n}(\omega) + 1)}\hat{A}(\omega), \tag{2.34}$$

$$L_\omega^- = \sqrt{\Gamma\bar{n}(\omega)}\hat{A}(\omega), \tag{2.35}$$

After discussing some of the open quantum systems theory, it is always helpful to solve an example. Below, we will solve the thermalisation of a single-mode harmonic oscillator, whose solution will be used later in this work's results.

2.2.1.1 Example: single-mode harmonic oscillator coupled to a thermal bath

Consider the system is a single-mode harmonic oscillator $\hat{H}_S = \omega_0 \hat{a}^\dagger \hat{a}$ with $\hat{a}$ and $\hat{a}^\dagger$ being its annihilation and creation operators, respectively. As we discussed before, this system is coupled to a thermal bath of temperature β, and the relaxation process is described by Eq. (2.33).

Let's find out the jump operators of this case. Usually, the Hermitian operator of the interaction acting on the system subspace is defined as $\hat{A} = \hat{a} + \hat{a}^\dagger$. To decompose $\hat{A}$ into $\hat{A}(\omega)$, we must find out the eigenstates of $\hat{H}_S$ and the frequencies at which this interaction operator makes transitions in our systems. The eigenstates of the harmonic oscillator are Fock states, $\hat{H}_S \left|n\right\rangle = \lambda_n \left|n\right\rangle$ with $\lambda_n = n\omega_0$. For these states, the creation operator raises one energy level ($\hat{a}^\dagger \left|n\right\rangle = \sqrt{n+1} \left|n+1\right\rangle$) and the annihilation operator lowers one energy level ($\hat{a} \left|n\right\rangle = \sqrt{n} \left|n-1\right\rangle$) Using that the energy difference is given by $\lambda_m - \lambda_n = (m-n)\omega_0$, the possible transition frequencies are $+\omega_0$ or $-\omega_0$. Then, the terms of the decomposition of $\hat{A}$ according to Eq. (2.10) becomes

$$\hat{A}(+\omega_0) = \sum_m \left|m-1\right\rangle \left\langle m-1\right| \hat{A} \left|m\right\rangle \left\langle m\right| = \hat{a}, \tag{2.36}$$

$$\hat{A}(-\omega_0) = \sum_m \left|m+1\right\rangle \left\langle m+1\right| \hat{A} \left|m\right\rangle \left\langle m\right| = \hat{a}^\dagger. \tag{2.37}$$

As a result, the jump operators of this thermal relaxation are

$$L_\omega^+ = \sqrt{\Gamma(\bar{n}(\omega)+1)}\hat{a}, \tag{2.38}$$

$$L_\omega^- = \sqrt{\Gamma\bar{n}(\omega)}\hat{a}^\dagger. \tag{2.39}$$

This example is fundamentally related to Gaussian states, which will be introduced in Sec. 4.2. In Sec. 4.2.1, we will analytically solve this master equation.

3 QUANTUM THERMODYNAMICS

3.1 Quantum Correlations

3.2 Thermodynamic variables in classical and quantum regime

3.3 Equilibrium and Nonequilibrium States

3.4 Relaxation, Entropy Production and Irreversibility

3.4.1 Non-Markovianity

Up to this point, we have discussed how to describe Markovian relaxation of open quantum systems. However, the Born-Markov approximation is not valid for many processes of open quantum systems, for example, with strong coupling cite, structured reservoirs (5, 6), or large initial system-environment correlations cite. In these cases, memory effects become relevant for the time evolution of the system, a phenomenon known as non-Markovian dynamics. There are a variety of ways to measure and witness non-Markovianity. For example, in information theory, integrating the trace distance in time can quantify the information backflow (7). Another example is analysing the divisibility of the maps that generate the system's dynamics (3). In this work, we are interested in a witness of non-Markovianity that connects the information and thermodynamic analysis: the entropy production (6, 8). The entropy production rate is defined as

$$\sigma(\hat{\rho}) = -\frac{d}{dt} D[\hat{\rho}(t)||\rho(0)], \tag{3.1}$$

where D is the relative entropy (also called Kullback-Leibler divergence) that measures the distinguishability of a state evolved in time $\rho(t)$ with its initial form $\rho(0)$ (9). It is known from the Second Law of thermodynamics that entropy cannot be destroyed, meaning that $\sigma \geq 0$. For example, in a thermal relaxation process, the entropy production is always positive while being monotonically decreasing because less entropy is produced when approaching equilibrium. However, it was recently found that the entropy production of a non-Markovian system can become negative for some time intervals (6, 8), meaning that entropy is being destroyed, which violates the Second Law. The missing piece of this interpretation is considering the environment. Using the interpretation that entropy is related to lack of information, when the environment is losing information to the system, its entropy must be increasing, compensating for the entropy decrease of the system while conserving the total entropy. This not only provides a satisfactory explanation but also indicates that a thermodynamic quantity, such as entropy production, can be used as a witness of non-Markovianity.

3.4.2 Thermal Kinematics

4 FROM DISCRETE TO CONTINUOUS VARIABLE QUANTUM SYSTEMS

Nature exhibits a variety of physical phenomena, some of which are better described by discrete-variable models, while others are naturally represented by continuous-variable models. Quantum mechanics is formulated in both discrete and continuous regimes, that complements each other in explaining physical reality. Finding the analogous phenomena in discrete and continuous-variable systems not only brings to light fundamental physics but also enriches potential technological applications. Motivated by these considerations, this chapter explores the thermal relaxation asymmetry in both discrete and continuous-variable quantum systems.

Since the main goal of this work is to understand how quantum resources affect heating and cooling asymmetry, we focus on specific models that show these properties. In the discrete-variable scenario, we consider two interacting qubits coupled to a global bosonic bath, which allows us to investigate the consequences of quantum correlations and local non-Markovian dynamics in thermal relaxation. In the continuous-variable scenario, we focus on Gaussian states due to their relevance in quantum theory and experimental accessibility. The quantum resources explored in this case are caused by the displacement and squeezing of thermal states.

This chapter discusses the methods and results from this work. The two interacting qubits case is discussed in Sec. 4.1. In Sec. 4.2.1, we present the necessary theory of Gaussian systems as well as our results for the thermal relaxation asymmetry of thermal (Sec. 4.2.1.1), displaced (Sec. 4.2.1.2), and squeezed (Sec. 4.2.1.3) Gaussian states.

4.1 Thermal Relaxation of Two Interacting Qubits

In this section, we discuss the effects of quantum correlations in the thermal relaxation of two interacting qubits. The model we use for this interaction is part of the family of Hamiltonians described by

$$H_S = -h\sum_{j=1}^{N}\sigma_j^z - g\sum_{j=1}^{N-1}[(1+\gamma)\sigma_j^x\sigma_{j+1}^x + (1-\gamma)\sigma_j^y\sigma_{j+1}^y + \Delta\sigma_j^z\sigma_{j+1}^z], \tag{4.1}$$

where N is the number of qubits, σ_j^x, σ_j^y, and σ_j^z are the Pauli matrices for the jth qubit, g is the coupling strength between the qubits, while γ and Δ creates anisotropies in the system. In all of our results and discussions, we consider $\hbar = 1$ and the Boltzmann constant $k_B = 1$. Depending on the values of γ and Δ, we can obtain different models (10), as presented in Tab. 1. An important aspect of this family of Hamiltonians is the possibility of production different correlations between the qubits, which enables our studies on quantum resources in thermal relaxation asymmetry.

Table 1 – Models and Parameters

Parameters	Models
$\Delta = 1,\ \gamma = 0$	XXX
$\Delta \neq 0,\ \gamma = 0$	XXZ
$\Delta \neq 0,\ -1 < \gamma < 1$	XYZ
$\Delta = 0,\ \gamma = 0$	XX
$\Delta = 0,\ -1 < \gamma < 1$	XY
$\Delta = 0,\ \gamma = \pm 1$	Transverse Field Ising (TFI)

We focus on only two qubits ($N = 0$) coupled to a global bath as shown in Fig. 1. Both qubits have an energy gap of $h = \frac{\omega_0}{2}$ with one of them heating up from the temperature T_c while the other is cooling down from the temperature T_h, and the bath is at an intermediate temperature T_w. To genuinely heat up or cool down the qubits, they must be thermal states, allowing the definition of a physical temperature. If we added off-diagonal elements to a qubit's density matrix, it would only be possible to define an effective temperature, making the terms "heating" and "cooling" inaccurate. Therefore, we introduce initial global coherences while keeping the reduced state of each qubit thermal throughout the dynamics, resulting in the density matrix

$$\rho = \rho_c \otimes \rho_h + \chi, \tag{4.2}$$

where ρ_c and ρ_h are the thermal states of the cold and hot qubits, respectively, and the global coherences are given by $\chi = e\,|01\rangle\langle 10| + e^*\,|10\rangle\langle 01| + f\,|00\rangle\langle 11| + f^*\,|11\rangle\langle 00|$ in the computational basis and has $\mathrm{Tr}_c[\chi] = \mathrm{Tr}_h[\chi] = 0$. The density matrix is always positive semi-definite, imposing constraints on the values of the coherence parameters $|e|$ and $|f|$. By diagonalizing ρ, we obtain that the maximum value for $|e|$ and $|f|$ is $1/\mathcal{Z}_c\mathcal{Z}_h$, with $\mathcal{Z}$ being the partition function. The detailed calculation can be found in apx:max-coherence.

The last step before discussing our results is to define T_c and T_h. To make a fair comparison between heating and cooling, we must find thermodynamically equidistant initial states from equilibrium. Given that the asymmetry only appears for far-from-equilibrium systems, we quantify this distance using the nonequilibrium free energy (F_{neq}

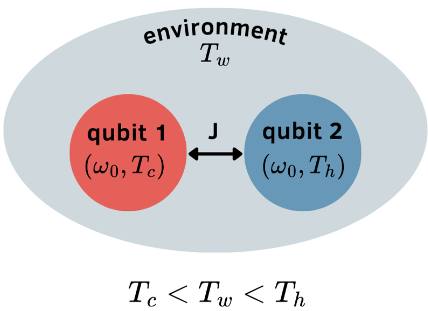

Figure 2 – Model for two interacting qubits with a global bath.

defined by Eq.) (11). Additionally, because F_{neq} is directly related to the relative entropy (defined by Eq.), initial states that are thermodynamically equidistant from equilibrium also have the same relative entropy from the steady state of the dynamics. In our case, we compare the cold and hot qubits' initial states with their local steady state (ρ_{ss}^{q}),

$$D[\rho_c||\rho_{ss}^{q}] = D[\rho_h||\rho_{ss}^{q}]. \tag{4.3}$$

By definition, a steady state does not change with time, which can be found by solving $\mathcal{L}(\rho_{ss}^{G}) = 0$, where ρ_{ss}^{G} is the global steady state, and $\mathcal{L}$ is the generator of the dynamics as discussed in Chap. 2. However, from Sec. 2.2.1, we know that the steady state of a Davies map is always a Gibbs state,

$$\rho_{ss}^{G} = \frac{e^{-\beta_w H_S}}{\mathcal{Z}_S}, \tag{4.4}$$

where $\beta_w = 1/T_w$, H_S is the system Hamiltonian (Eq. (4.1)) and $\mathcal{Z}_S$ is the partition function. Then, we can trace out one of the qubits to find ρ_{ss}^{q}. Because in a global bath both qubits equilibrate to the same temperature, it makes no difference which one we trace out. However, it should be emphasised that, even though the global system thermalises to the bath's temperature, the qubit's temperature of equilibrium is different. This happens because the interaction between the qubits generates correlations that are erased by the partial trace.

Now that we discussed the interaction model between the qubits and the structure of the initial state, we can solve the master equation given by the Davies maps presented in Chap. 2. With the local density matrices, we compute the Quantum Fisher Information defined in Eq.. Then, we obtain the thermal kinematics variables presented in Chapter 3 to analyse the thermal relaxation asymmetry of our system.

4.1.1 Results for the XX Model

This section presents the results regarding the heating and cooling asymmetry for the XX model ($\gamma = 0$ and $\Delta = 0$). From the Hamiltonian in Eq. (4.1), used $h = \omega_0/2$ and redefined $g = 2J$. For this model, the qubit local steady state is

$$\rho_{ss}^{q} = \frac{1}{\mathcal{Z}_q}\begin{bmatrix} e^{\beta_w\omega_0} + \cosh(\beta_w J) & 0 \\ 0 & e^{-\beta_w\omega_0} + \cosh(\beta_w J) \end{bmatrix}, \tag{4.5}$$

where $\mathcal{Z}_q = 2\cosh(\beta_f\omega_0) + 2\cosh(\beta_f J)$. The local temperature of this state is

$$\beta_{qubit}^{f} = \frac{1}{\omega_0}\ln\left(\frac{e^{\beta_w\omega_0} + \cosh(\beta_w J)}{e^{-\beta_w\omega_0} + \cosh(\beta_w J)}\right), \tag{4.6}$$

that is different from the bath's temperature. Using this result, the initial temperatures T_c and T_h are obtained numerically from Eq. (4.3), ensuring a fair comparison between heating and cooling.

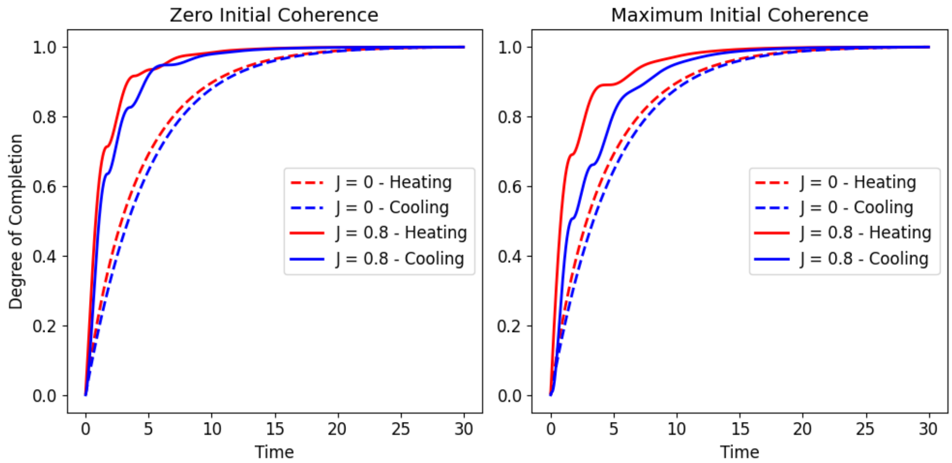

Figure 3 – Degree of completion for XX-model. Each line correspond to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

With these temperatures defined, we can compare the dynamics of the heating and cooling qubits. In our results, heating and cooling are represented in red and blue, respectively. Dashed lines indicate the non-interaction case ($J = 0$), while solid lines represent the interaction case ($J = 0.8$). The degree of completion (Eq.) is shown in Fig. 3 for the case without initial coherence (left) and the case with maximum initial coherence (right). Given that the red lines are always above their corresponding blue lines, we conclude that heating is faster than cooling. Using this same idea, we see that the interaction between qubits allows a faster relaxation. Furthermore, only for the interaction case, the initial coherence seems to affect the intermediary dynamics by delaying cooling, which increases the asymmetry while keeping the total relaxation time unchanged. These same observations can be obtained from the statistical velocity (Eq.) shown in Fig. 4.

In addition to our previous observations, note that both the statistical velocity and the degree of completion have oscillations. To understand better its cause, we calculated the entropy production rate (Π_ρ defined by Eq.) that is shown in Fig. 5. Only for the interaction case, the entropy production rate becomes negative for some time intervals, which is a witness of non-Markovian dynamics, where information from the environment returns to the systems (6). Another interesting observation is that heating has a bigger initial entropy production rate than cooling, which corroborates with the hypothesis that heating is faster because its entropy increases faster than cooling decreases its energy (12).

Another interesting behaviour of the interaction case is that it generates correlations that are signalised by the mutual information defined in Eq.. Until this moment, we used an interaction that do not create entanglement between the qubits despite of generating correlations as shown in Fig. 6. Now, let's compare what happens to the asymmetry when we have the entanglement formation. For $J = 1.6$ we see in Fig. 7 that the entangle-

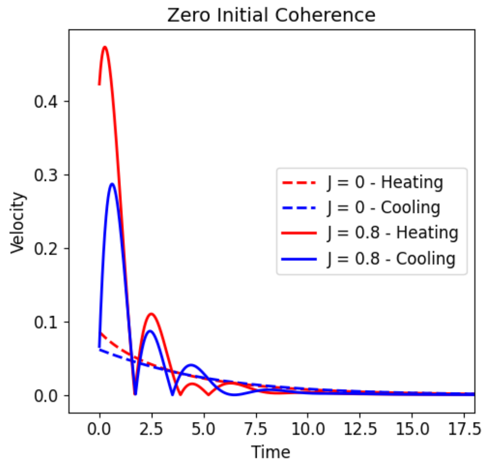

Figure 4 – Statistical velocity for XX-model. Each line correspond to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

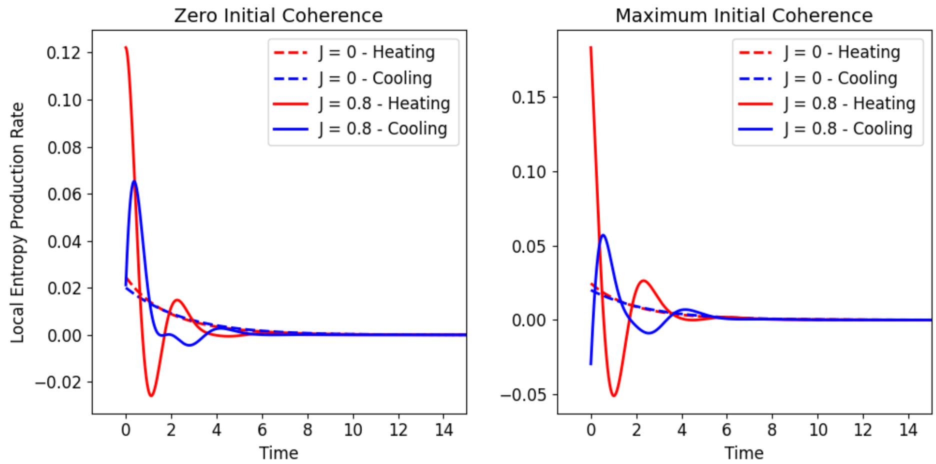

Figure 5 – Entropy production rate for the XX-model. Each line corresponds to cooling (blue) or heating (red) processes with (solid) or without (dashed) interaction between qubits.

ment of formation (Eq.) increases with time until reaching a plateau in the equilibrium. It is interesting that the presence of maximum initial coherence delays the generation of entanglement. In Fig. 8 we see that the presence of entanglement (solid lines) makes the thermal relaxation more symmetric and slower when compared to the $J = 0.8$ case (dashed lines), but still is faster than the non-interacting case. Again the initial coherence do not change the overall thermalisation, but affects the beginning of the dynamics by making a small increase of the asymmetry.

4.2 Gaussian States

Continuous-variable (CV) systems are described by observables with continuous spectra, typically associated with infinite-dimensional Hilbert spaces. An important example is the quantised electromagnetic field, whose modes behave as independent harmonic oscillators. Despite being described by continuous variables, the energy of each mode is quantised in equally spaced levels, with transitions occurring through the absorption or emission of energy quanta known as photons. For simplicity, let us consider an electromagnetic field inside an optical cavity, where only N relevant modes are taken into account. In this case, the Hamiltonian can be written as the sum of independent modes, $H = \sum_{k=1}^{N} H_k$, with each mode described by the harmonic oscillator Hamiltonian,

$$H_k = \hbar\omega_k \left(\hat{a}_k^{\dagger}\hat{a}_k + \frac{1}{2}\right). \tag{4.7}$$

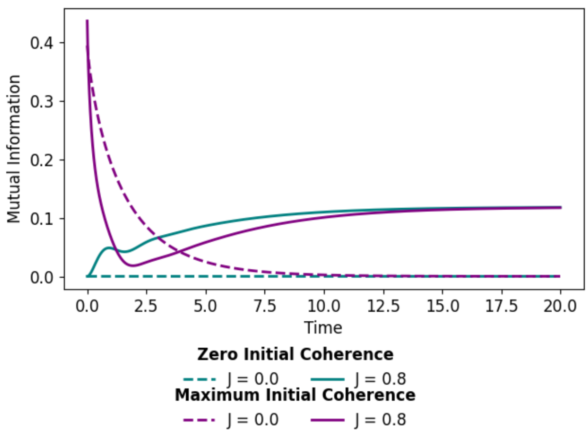

Figure 6 – Mutual information as a function of time for the zero (teal) and maximum (purple) initial coherence with $J = 0$ (dashed) and $J = 0.8$ (solid) cases.

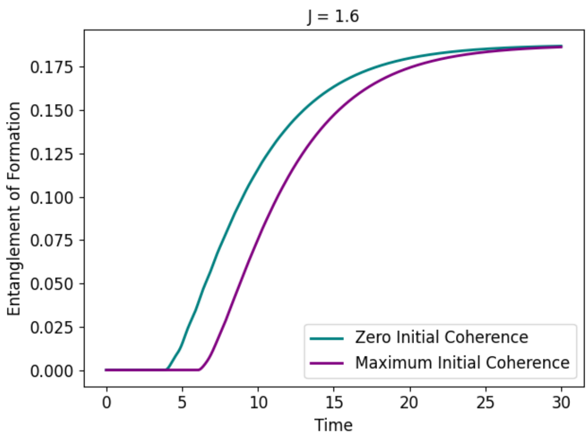

Figure 7 – Entanglement of formation as a function of time for the zero (teal) and maximum (purple) initial coherence with $J = 1.6$.

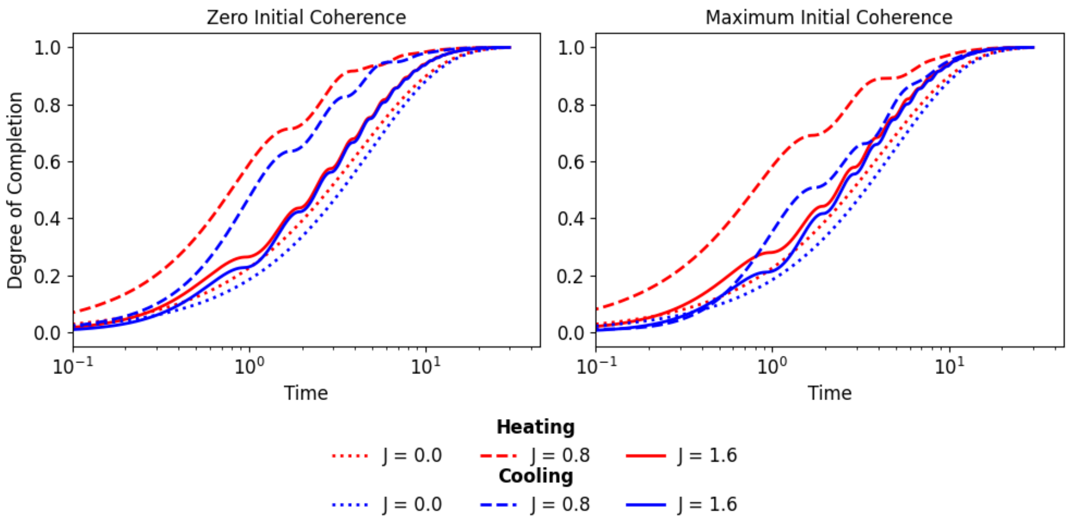

Figure 8 – Degree of completion for heating (red) and cooling (blue) states with zero (left) or maximum (right) initial coherence. The interaction parameters used are $J = 0$ (dotted), $J = 0.8$ (dashed), and $J = 1.6$ (solid).

Here, $\hat{a}_k^\dagger$ and $\hat{a}_k$ are the creation and annihilation operators of the mode k. The origin of these names is that they create or annihilate a photon with wave vector $\mathbf{k}$, changing the mode energy by one quantum. $\hat{a}_k^\dagger$ and $\hat{a}_k$ satisfy the following commutation relations

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \tag{4.8}$$

$$[\hat{a}_i^\dagger, \hat{a}_j^\dagger] = [\hat{a}_i, \hat{a}_j] = 0. \tag{4.9}$$

The quantum harmonic oscillator can equivalently be described in terms of the position and momentum operators, also called quadratures,

$$\hat{q}_k = \frac{1}{\sqrt{2}}(\hat{a}_k^\dagger + \hat{a}_k) \tag{4.10}$$

$$\hat{p}_k = \frac{i}{\sqrt{2}}(\hat{a}_k^\dagger - \hat{a}_k). \tag{4.11}$$

For an N-mode systems, the quadratures can be grouped into a vector $\hat{R} = (\hat{R}_1, \dots, \hat{R}_N)^T = (\hat{q}_1, \hat{p}_1, ..., \hat{q}_N, \hat{p}_N)^T$, allowing the commutation relations to be condensed in $[\hat{R}_i, \hat{R}_j] = i\Omega_{ij}$, where

$$\Omega = \bigoplus_{k=1}^{N} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \tag{4.12}$$

is the N-mode symplectic form. The quadratures operators define the phase space $\mathcal{P} = (\mathbb{R}^{2N}, \Omega)$, a real 2N-dimensional space provided with the symplectic structure Ω. This structure encodes the Heisenberg uncertainty principle. As a consequence, quantum states are represented as phase-space regions instead of single points like in classical mechanics.

The eigenstates of each Hamiltonian H_k in Eq. (4.7) are the Fock states $|n\rangle_k$, where n denotes the number of photons in mode k. As a consequence, the basis for the full multimode Hamiltonian is given by the tensor product $|n\rangle_{\mathbf{k}} = |n\rangle_1 \otimes |n\rangle_2 \otimes \cdots \otimes |n\rangle_N$. Each Fock basis is bounded from below by the vacuum state $|0\rangle_k$, defined by $\hat{a}_k |0\rangle_k = 0$. Although the Fock representation is important for describing quantised fields, it becomes less convenient when dealing with states containing a large number of photons, such as laser fields. In these cases, an alternative representation is to use coherent states. A single-mode coherent state can be written as a linear combination of Fock states,

$$|\alpha\rangle_k = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle_k . \tag{4.13}$$

Experimentally, coherent states may be generated by applying the Weyl operator, also known as the displacement operator, to the vacuum state,

$$|\alpha\rangle_k = \hat{D}_k(\alpha) |0\rangle_k , \tag{4.14}$$

where the Weyl operator is

$$\hat{D}_k(\alpha) = e^{\alpha \hat{a}_k^\dagger - \alpha^* \hat{a}_k}, \tag{4.15}$$

with $\alpha \in \mathbb{C}$ being the displacement parameter. For the N-mode case, the displacement operator can be written in terms of the canonical operator vector $\hat{R}$ as

$$\hat{D}(\mathbf{r}) = e^{\hat{R}^T \Omega \mathbf{r}}, \tag{4.16}$$

where $\mathbf{r} = (q'_1, p'_1, \ldots, q'_N, p'_N)$ is a vector in phase-space $\mathcal{P}$ specifying the displacement applied for each quadrature.

Until this moment, we have been using density matrices to describe our states. However, the density operators of CV systems are infinite-dimensional, making this approach mathematically complicated. An alternative is to work with equivalent phase-space representations. The characteristic function is a multivariable function obtained through the Fourier-Weyl transform of the density operator (13), being written as

$$\chi_\rho(\mathbf{r}) = \mathrm{Tr}[\rho \hat{D}(\mathbf{r})]. \tag{4.17}$$

In this work, we will consider only the symmetric-ordered characteristic function defined above; other orderings can be found in Refs. (13, 14). Although χ_ρ fully characterises a quantum state, it is still expressed in terms of displacement variables rather than quadrature operators. The full transition to phase-space can be done by taking the Fourier transform of the characteristic function, resulting in the Wigner quasiprobability distribution,

$$W_\rho(\mathbf{q}, \mathbf{p}) = \frac{1}{\pi^N} \int_{\mathbb{R}^N} \langle \mathbf{q} + \mathbf{q}' | \rho | \mathbf{q} - \mathbf{q}' \rangle e^{2i\mathbf{q}' \cdot \mathbf{p}} d^N \mathbf{q}'. \tag{4.18}$$

The derivation of the Wigner distribution is in apx:wigner. Here, $\mathbf{q}$ and $\mathbf{p}$ are N-dimensional vectors indicating the coordinates of a point in the phase-space. The vector $\mathbf{x}$ is a symmetric displacement around $\mathbf{q}$, being the integration variable and $\hat{q}_k \mathbf{x} = q_k \mathbf{x}$. The Wigner distribution is called a quasiprobability because, unlike classical probability distributions, it may take negative values. Still, its marginalisation reproduces the correct probability distribution associated with the quadrature observables (14).

Among the different classes of continuous-variable states, Gaussian states play an important role due to their mathematical simplicity and experimental relevance. As the name suggests, the Wigner distribution associated with these states is a Gaussian function in phase space. For a single variable, a Gaussian function can be written as

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right), \tag{4.19}$$

where $\mu = \langle x \rangle$ is the mean value (first moment) and $\sigma^2 = \langle (x - \langle x \rangle)^2 \rangle$ is the variance (second moment). These two quantities fully characterise a Gaussian distribution. Similarly, multimode Gaussian states are fully characterised by the first and second moments of the quadrature operators. The first moments are grouped into the mean vector $\mathbf{v}$, whose elements are

$$v_j = \langle \hat{R}_j \rangle_\rho, \tag{4.20}$$

while the second moments are written in the covariance matrix Θ, with elements

$$\Theta_{ij} = \frac{1}{2} \langle \hat{R}_i \hat{R}_j + \hat{R}_j \hat{R}_i \rangle_\rho - \langle \hat{R}_i \rangle_\rho \langle \hat{R}_j \rangle_\rho, \tag{4.21}$$

where $\langle \hat{O} \rangle_\rho \equiv \mathrm{Tr}[\rho \hat{O}]$. With these definitions, the Wigner distribution becomes

$$W_\rho(\mathbf{x}) = \frac{1}{\pi^N \sqrt{|\Theta|}} \exp\left(-(\mathbf{x}-\mathbf{v})^T \Theta^{-1} (\mathbf{x}-\mathbf{v})\right), \tag{4.22}$$

where $\mathbf{x} \in \mathbb{R}^{2N}$ and $|\Theta| = \det(\Theta)$. Important examples of Gaussian states are coherent states, thermal states, and the vacuum state.

In addition to characterising Gaussian states, we are interested in the operations that act upon them. Although general quantum operations may destroy Gaussianity, Gaussian operations preserve the Gaussian form of the state in phase space. In this work, we focus specifically on displacement operations, defined in Eq. (4.15), and squeezing operations.

The displacement operator, as the name suggests, shifts the whole Gaussian state without deforming the distribution. In other terms, it varies the mean vector while keeping the covariance matrix unchanged. On the other hand, the squeezing operator for a single mode is defined as

$$\hat{S}_k(z) = e^{-\frac{1}{2}(z^* \hat{a}_k - z \hat{a}_k^\dagger)}, \tag{4.23}$$

where $z = re^{i\phi_s}$ with r being the squeezing parameter and ϕ_s being the squeezing phase responsible to rotate the state. The squeezing operator deforms the state in the phase

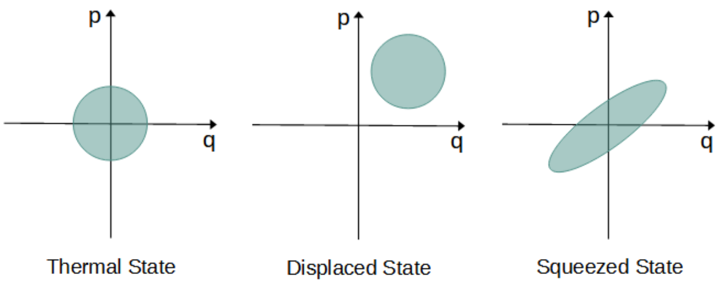

Figure 9 – Illustration of a single-mode thermal state, displaced coherent state, and squeezed coherent state in the phase space.

space by compressing one quadrature. As a consequence of the uncertainty principle, the other quadrature will expand. Mathematically, this means that squeezing operations only affect the covariance matrix, preserving the mean vector. An illustration of how these operations affect a Gaussian state is shown in Figure 9. We will not use the multimode squeezing operator in this work, but its definitions can be found in Refs. (13, 14).

From a thermodynamic perspective, squeezing and displacement operations add energy to the system, bringing it to the far-from-equilibrium regime (15). One way to quantify this energy gain is by a quantity called ergotropy, which is defined as the maximum amount of energy that can be extracted from a state via cyclic unitary operations (16),

$$\mathcal{E}(\hat{\rho}, \hat{H}) = E(\hat{\rho}) - E(\hat{\pi}), \tag{4.24}$$

where $E(\hat{\rho}) = \mathrm{Tr}[\hat{\rho}\hat{H}]$ is total energy of the system and $E(\hat{\pi}) = \min_{\hat{U}} \mathrm{Tr}[\hat{U}\hat{\rho}\hat{U}^\dagger \hat{H}]$ is the energy of the correspondent state that no work can be extracted via unitary operations, being called *passive state* and represented by $\hat{\pi}$. For Gaussian states, all passive states are thermal. That said, the ergotropy for a Gaussian state W with mean vector $\mathbf{v}$ and covariance matrix Θ is defined as

$$\mathcal{E}(W) = \omega f(\beta_\pi) K[W||W_\pi], \tag{4.25}$$

where W_π is the Wigner function of the passive state, $f(\beta_\pi) = \sqrt{|\Theta|}$ being β_π the inverse temperature of the passive state, and K is the Wigner relative entropy defined as

$$K[W_1||W_2] = -1 + \frac{1}{2}\left[(\mathbf{v}_1 - \mathbf{v}_2)^\dagger \Theta_2^{-1}(\mathbf{v}_1 - \mathbf{v}_2) + \mathrm{Tr}(\Theta_2^{-1}\Theta_1) + \ln\frac{|\Theta_2|}{|\Theta_1|}\right]. \tag{4.26}$$

Recently, it was shown in Ref. (16) that the relaxation of squeezed states is faster than that of a displaced state with the same initial ergotropy. These results inspired the study of how displacement and squeezing may affect the thermal relaxation asymmetry. In Section 4.2.1, we will see how Gaussian states evolve in time and will redefine the thermal kinematics variables for them.

4.2.1 Thermal Relaxation of Gaussian States

As discussed in Chap. 2, the dynamics of an open system can be described by the Lindblad master equation. In our case, we will consider a single bosonic mode (Eq. (4.7) with $k = 1$) interacting with a thermal bath, which has already been derived in Sec. 2.2.1.1,

$$\frac{d}{dt}\hat{\rho} = -i[\hat{H}, \hat{\rho}] + \Gamma(\bar{n}+1)\mathcal{D}_{\alpha}(\hat{\rho}) + \Gamma\bar{n}\mathcal{D}_{\alpha^\dagger}(\hat{\rho}), \tag{4.27}$$

where Γ is the dissipation rate, $\bar{n} = 1/(e^{\beta\omega} - 1)$ is the Bose distribution, and $\mathcal{D}_L(\hat{\rho}) = L\hat{\rho}L^\dagger - \frac{1}{2}\{L^\dagger L, \hat{\rho}\}$ is the dissipator. As elaborated in the previous section, it is convenient to use the phase-space representation instead of the density matrix. For this reason, we use Eq. (4.27) in $\frac{d}{dt}\langle A\rangle_\rho = \mathrm{Tr}[A\frac{d}{dt}\rho]$ to find the differential equations for the mean vector and covariance matrix.

The mean vector of a single mode has the form $\mathbf{v}_t = (\langle\hat{a}\rangle, \langle\hat{a}^\dagger\rangle)^T$. To find its time derivative, we compute

$$\frac{d}{dt}\langle\hat{a}\rangle = -\left(i\omega + \frac{\Gamma}{2}\right)\langle\hat{a}\rangle \tag{4.28}$$

$$\frac{d}{dt}\langle\hat{a}^\dagger\rangle = \left(i\omega - \frac{\Gamma}{2}\right)\langle\hat{a}^\dagger\rangle, \tag{4.29}$$

which gives the differential equation

$$\dot{\mathbf{v}}_t = \Lambda\mathbf{v}_t, \tag{4.30}$$

where

$$\Lambda = -\frac{1}{2}\begin{bmatrix}\Gamma + 2i\omega & 0 \\ 0 & \Gamma - 2i\omega\end{bmatrix}. \tag{4.31}$$

The covariance matrix of a single mode is structured as

$$\Theta_t = \begin{bmatrix}\frac{1}{2}(\langle\hat{a}^\dagger\hat{a}\rangle + \langle\hat{a}\hat{a}^\dagger\rangle) - \langle\hat{a}^\dagger\rangle\langle\hat{a}\rangle & \langle\hat{a}^2\rangle - \langle\hat{a}\rangle^2 \\ \langle\hat{a}^{\dagger 2}\rangle - \langle\hat{a}^\dagger\rangle^2 & \frac{1}{2}(\langle\hat{a}^\dagger\hat{a}\rangle + \langle\hat{a}\hat{a}^\dagger\rangle) - \langle\hat{a}^\dagger\rangle\langle\hat{a}\rangle\end{bmatrix}. \tag{4.32}$$

To take its time derivative, we first compute

$$\frac{d}{dt}\langle\hat{a}\hat{a}^\dagger\rangle = -\Gamma(\langle\hat{a}\hat{a}^\dagger\rangle - \bar{n} - 1), \tag{4.33}$$

$$\frac{d}{dt}\langle\hat{a}^\dagger\hat{a}\rangle = -\Gamma(\langle\hat{a}\hat{a}^\dagger\rangle - \bar{n}), \tag{4.34}$$

$$\frac{d}{dt}\langle\hat{a}^2\rangle = -(2i\omega + \Gamma)\langle\hat{a}^2\rangle, \tag{4.35}$$

$$\frac{d}{dt}\langle\hat{a}^{\dagger 2}\rangle = (2i\omega - \Gamma)\langle\hat{a}^{\dagger 2}\rangle, \tag{4.36}$$

to obtain the following differential equation called the Lyapunov Equation,

$$\dot{\Theta}_t = \Lambda\Theta + \Theta\Lambda^\dagger + F, \tag{4.37}$$

where $F = \Gamma f(\beta)\mathbb{I}$. Therefore, by solving Eqs. (4.30) and (4.37), we find that the time evolution of a single-mode Gaussian state is given by

$$\mathbf{v}_t = e^{-\Gamma t}(\langle a\rangle_0 \, e^{-i\omega t}, \langle a^\dagger\rangle_0 \, e^{i\omega t})^T \tag{4.38}$$
$$\Theta_t^{11} = \Theta_t^{22} = c_0 e^{-\Gamma t} + f(\beta), \tag{4.39}$$
$$\Theta_t^{12} = (\Theta_t^{21})^* = d_0 e^{-(\Gamma+2i\omega)t}, \tag{4.40}$$

where c_0, d_0, $\langle a\rangle_0$, and $\langle a^\dagger\rangle_0$ are constants obtained from the initial conditions.

After finding the dynamics of a single-mode Gaussian state, we are now ready to investigate the heating and cooling asymmetry in these systems. As discussed in Chapter 3, Fisher Information is the essential quantity for the thermal kinematics approach of thermal relaxation asymmetry. For single-mode Gaussian systems, we can find the Wigner-Fisher Information by making a Taylor expansion of $K[W_{t+dt}||W_t]$ (the derivation is in apx:wigner-fisher), resulting in

$$\mathcal{I}_W(t) = \dot{\vec{v}}_t^\dagger \Theta_t^{-1} \dot{\vec{v}}_t + \mathrm{Tr}[(\Theta_t^{-1}\dot{\Theta}_t)^2], \tag{4.41}$$

Having introduced the necessary formalism, we will start discussing our results. Although the heating and cooling asymmetry for Gaussian thermal states has already been calculated with numerical methods (17), the following section presents an analytical treatment of this problem.

4.2.1.1 Asymmetry for Thermal States

In this section, we consider a Gaussian state initially in a thermal state with inverse temperature β_i interacting with a bath at inverse temperature β_f. The system's dynamics is described by a zero mean vector ($\mathbf{v}_t = 0$) and a covariance matrix equal to $\Theta_t = \Delta_\beta \mathbb{I}$, being $\Delta_\beta = [f(\beta_i) - f(\beta_f)]e^{-\Gamma t} + f(\beta_f)$ where Γ is the dissipation rate, t is the time and $f(\beta) = \bar{n} + 1/2$.

Similar to the qubits case, to find two thermodynamically equivalent initial states, we use the Wigner relative entropy defined in Eq. (4.26). For thermal states, $|\Theta| = (\Delta_\beta)^2$ and $\Theta^{-1} = \frac{1}{\Delta_\beta}\mathbb{I}$, being $\lim_{t\to\infty} \Delta_\beta = f(\beta_f)$ in equilibrium. Using this, we obtain that the Wigner relative entropy for a thermal state is

$$K[W_t||W_f] = -1 + \frac{\Delta_\beta}{f(\beta_f)} + \ln \frac{f(\beta_f)}{\Delta_\beta}. \tag{4.42}$$

The analytical expression for the Wigner-Fisher Information (defined in Eq. (4.41)) is

$$\mathcal{I}_W(t) = 2\left[\frac{\partial}{\partial t}\ln \Delta_\beta\right]^2, \tag{4.43}$$

the statistical velocity (defined in Eq.)

$$v_W(t) = \frac{\partial}{\partial t} \ln \Delta_\beta, \tag{4.44}$$

and the statistical length (defined in Eq.)

$$\mathcal{L}_W(t) = \ln\left(\frac{[f(\beta_i) - f(\beta_f)]e^{-\Gamma t} + f(\beta_f)}{f(\beta_i)}\right). \tag{4.45}$$

Note that the statistical length in equilibrium is

$$\mathcal{L}_W^{eq} \equiv \lim_{t\to\infty} \mathcal{L}_W(t) = \ln\left[\frac{f(\beta_f)}{f(\beta_i)}\right], \tag{4.46}$$

which means that, if there are two states with different initial temperatures that relaxate to the same final temperature, their statistical length will be different. Lets define $f_i \equiv f(\beta_f)/f(\beta_i)$ and $k = \exp(-\Gamma t)$, and use the expressions (4.45) and (4.46) to write the degree of completion as

$$\varphi(k) = \frac{\ln(k + (1-k)f_i)}{\ln(f_i)}. \tag{4.47}$$

Now, we can analyse the thermal relaxation asymmetry for thermal states. First, we can use the Jensen inequality for $f(x) = \ln(x)$ to find that

$$\ln(k + (1-k)f_i) \geq (1-k)\ln(f_i). \tag{4.48}$$

We know that $\beta_c > \beta_f > \beta_h$, which results in $f(\beta_c) < f(\beta_f) < f(\beta_h)$, leading to

$$1 > \frac{f(\beta_f)}{f(\beta_h)} = f_h \tag{4.49}$$

$$1 < \frac{f(\beta_f)}{f(\beta_c)} = f_c. \tag{4.50}$$

This means that $\ln(f_c) > 0$ and the heating degree of completion (φ_c) is

$$\varphi_c = \frac{\ln(k + (1-k)f_c)}{\ln(f_c)} \geq 1 - k. \tag{4.51}$$

On the other hand, $\ln(f_h) < 0$ and the cooling degree of completion (φ_h) is

$$\varphi_h = \frac{\ln(k + (1-k)f_h)}{\ln(f_h)} \leq 1 - k. \tag{4.52}$$

With this, we conclude that heating is always faster than cooling for Gaussian thermal states ($\varphi_c \geq \varphi_h$).

It is relevant to study whether the presence of thermal relaxation asymmetry in Gaussian systems is also explained by the difference in entropy production rate. Taking the negative of the time derivative of Wigner relative entropy (Eq. (4.42)), we find that the entropy production rate is

$$\Pi_W = \frac{\dot{\Delta}_\beta}{\Delta_\beta} - \frac{\dot{\Delta}_\beta}{f(\beta_f)}. \tag{4.53}$$

From this expression, we obtain Fig. 10 by using $K[W_c||W_f] = K[W_h||W_f] = 0.5$ and $T_f = 2$ (in natural units). Note that heating reaches zero before cooling, corroborating our conclusion that heating relaxes faster than cooling. Also, heating has an initially higher entropy production rate than cooling, which explains the asymmetry.

4.2.1.2 Asymmetry for Displaced States

After displacing a thermal state, the mean vector becomes $\vec{v}_t = (\mu_t, \mu_t^*)$ where $\mu_t = \mu e^{-(i\omega+\Gamma/2)t}$ while the covariance matrix remains unchanged, i.e., $\Theta_t = \Delta_\beta \mathbb{I}$. Using these results in the Wigner relative entropy (Eq. (4.26)), we find

$$K[W_t||W_f] = -1 + \frac{|\mu|^2}{f(\beta_f)} e^{-\Gamma t} + \frac{\Delta_\beta}{f(\beta_f)} + \ln \frac{f(\beta_f)}{\Delta_\beta}. \tag{4.54}$$

Term containing the displacement parameter (μ) is independent of the system's initial temperature, which means that, regardless of the initial temperature, applying a displacement operator to any thermal state results in the same shift in their Wigner relative entropy. Thus, for simplicity, we chose two equidistant thermal states and then applied the same amount of work on both (which is equivalent to using the same μ). As we did before, we could try to analytically calculate the thermal kinematics variables, but the Wigner-Fisher Information for a displaced state is

$$I_W(t) = \frac{2}{\Delta_\beta} \left[\frac{\dot{\Delta}_\beta^2}{\Delta_\beta} + \left(\omega^2 + \frac{\Gamma^2}{4} \right) |\mu|^2 e^{-\Gamma t} \right], \tag{4.55}$$

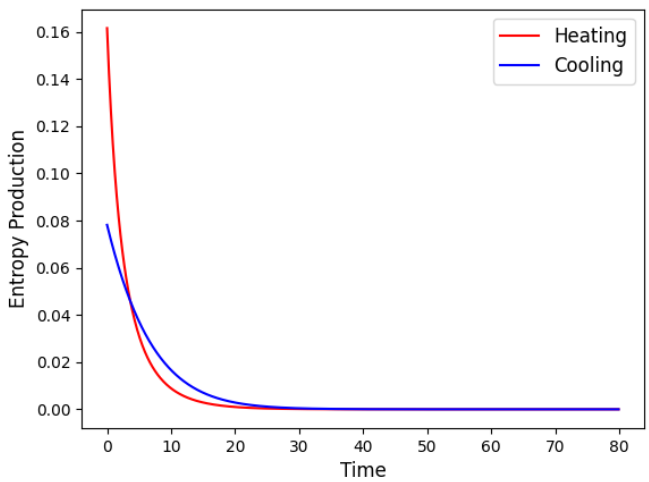

Figure 10 – Entropy production rate for heating (red) and cooling (blue) thermal states.

and, despite being possible to analytically find the statistical velocity and length, the complexity of the final expressions ultimately makes it difficult to extract useful information. Therefore, we use a numerical approach for this case. We used $\omega = 1$, $\Gamma = 0.1$, $T_f = 2$, and $K[W_c||W_f] = K[W_h||W_f] = 0.5$. Fig. 11 shows the degree of completion for different displacement parameters, with the thermal relaxation of W_c (in red) faster than that of W_h (in blue) for all the displacements.

Fig. 12 shows that the initial relative entropy increases with μ. This means that displacement operations move thermal states away from equilibrium. Another interesting aspect is that a greater distance from equilibrium leads to a higher Wigner Fisher Information, increasing the velocity as shown in Fig. 13.

To understand how displacement is affecting the relaxation, it is interesting to separate the passive contribution from the ergotropic contribution in the entropy production rate. The total entropy production rate for a displaced state is given by

$$\Pi_W = \frac{1}{f(\beta)}(\Gamma|\mu|^2 e^{-\Gamma t} - \dot{\Delta}_\beta) + \frac{\dot{\Delta}_\beta}{\Delta_\beta}. \tag{4.56}$$

This entropy production rate can also be rewritten as

$$\Pi_W = \dot{S}_{W_t} - \dot{S}_{W_\pi} + \Pi_{W_\pi} - \frac{\dot{\mathcal{E}}}{\omega f(\beta)}, \tag{4.57}$$

where $\dot{S}_W$ is the time derivative of Wigner entropy, Π_{W_π} is the entropy production for the passive state defined as

$$\Pi_{W_\pi} = \dot{S}_{W_t}\left(1 - \frac{f(\beta_\pi)}{f(\beta)}\right), \tag{4.58}$$

and $\dot{\mathcal{E}}$ is the ergotropic loss rate, that in this case is the time derivative of $\mathcal{E} = |\mu|^2 e^{-\Gamma t}$. Because $|\Theta_\pi| = |\Theta_t|$, we have $\dot{S}_{W_t} = \dot{S}_{W_\pi}$, which cancels out in Eq. (4.57). Fig. 14 shows

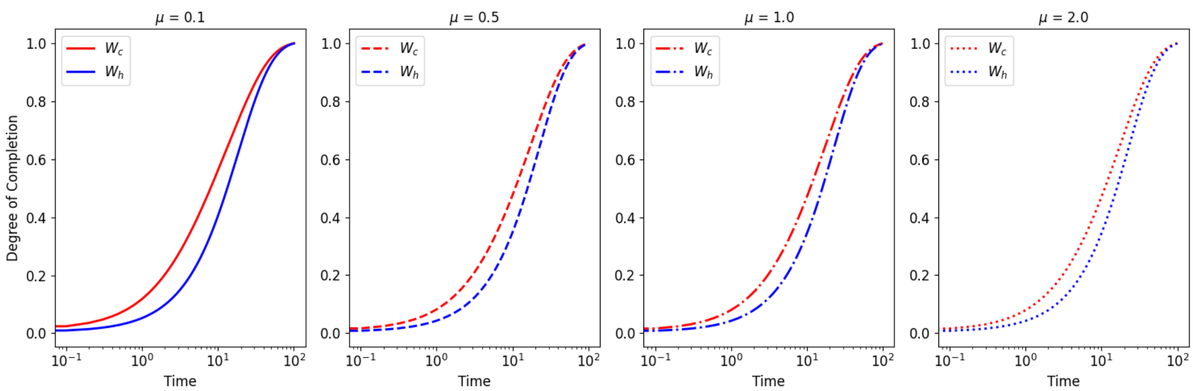

Figure 11 – Degree of completion for a displaced thermal state. Each graph corresponds to a different displacement parameter (μ).

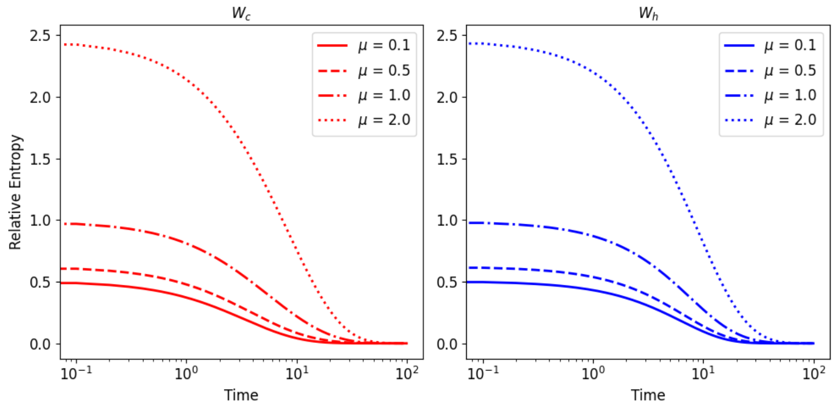

Figure 12 – Relative entropy of a displaced thermal state as a function of time. Each curve corresponds to a different displacement parameter (μ).

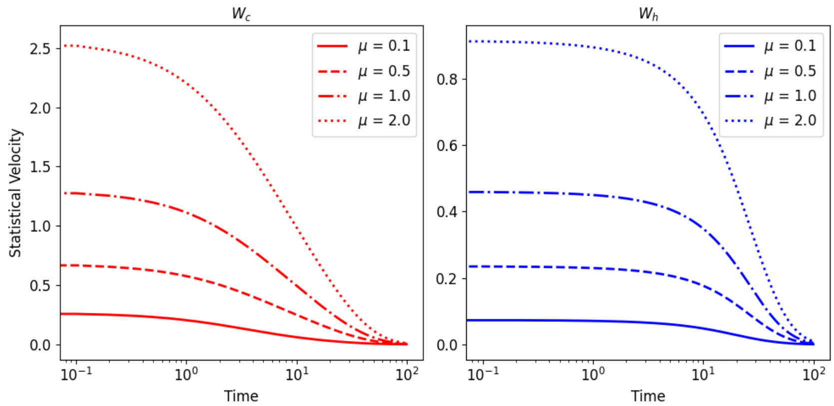

Figure 13 – Statistical velocity of a displaced thermal state as a function of time. Each curve corresponds to a different displacement parameter (μ).

the total entropy production rate (solid line), its passive contribution (dashed line), and its ergotropic loss rate contribution (dotted line). Each graph has a different displacement parameter μ, being W_c in red and W_h in blue.

In all the graphs, W_c starts with a higher total entropy production rate than W_h, which again explains why the thermal relaxation of W_c is faster. In addition, making greater displacements increases the ergotropic contribution while keeping the passive contribution unchanged. As a consequence, the total entropy production increases with the bigger μ. Another interesting result is that the ergotropic contribution is the same for W_c and W_h with same μ, being the passive contribution the generator of asymmetry and unaffected by the displacement operation.

4.2.1.3 Asymmetry for Squeezed States

In this section, we study the thermal relaxation asymmetry with squeezed states. Unlike the displacement operation, squeezing a thermal state preserves the zero mean vector and changes the covariance matrix to

$$\Theta_t = f(\beta_t)\begin{bmatrix} \cosh(2r_t) & -e^{i\phi_t}\sinh(2r_t) \\ -e^{-i\phi_t}\sinh(2r_t) & \cosh(2r_t) \end{bmatrix}, \tag{4.59}$$

where ϕ_t is the squeezing phase, the squeezing parameter is implicitly defined as

$$\cosh(2r_t) = \frac{1}{f(\beta_t)}[\Delta_\beta + 2f(\beta_i)e^{-\Gamma t}\sinh^2(r)], \tag{4.60}$$

and β_t is implicitly defined by

$$f(\beta_t) = \sqrt{\Delta_\beta^2 + 4f(\beta_i)f(\beta)e^{-2\Gamma t}(e^{\Gamma t}-1)\sinh^2(r)}. \tag{4.61}$$

With the information above, the relative entropy for a squeezed state is

$$K[W_t||W_f] = -1 + \frac{f(\beta_t)}{f(\beta)}\cosh(2r_t) + \ln\left(\frac{f(\beta)}{f(\beta_t)}\right). \tag{4.62}$$

From this expression, we can notice that squeezing thermal states with different temperatures do not preserve their equidistance, unlike the displacement operation case. As

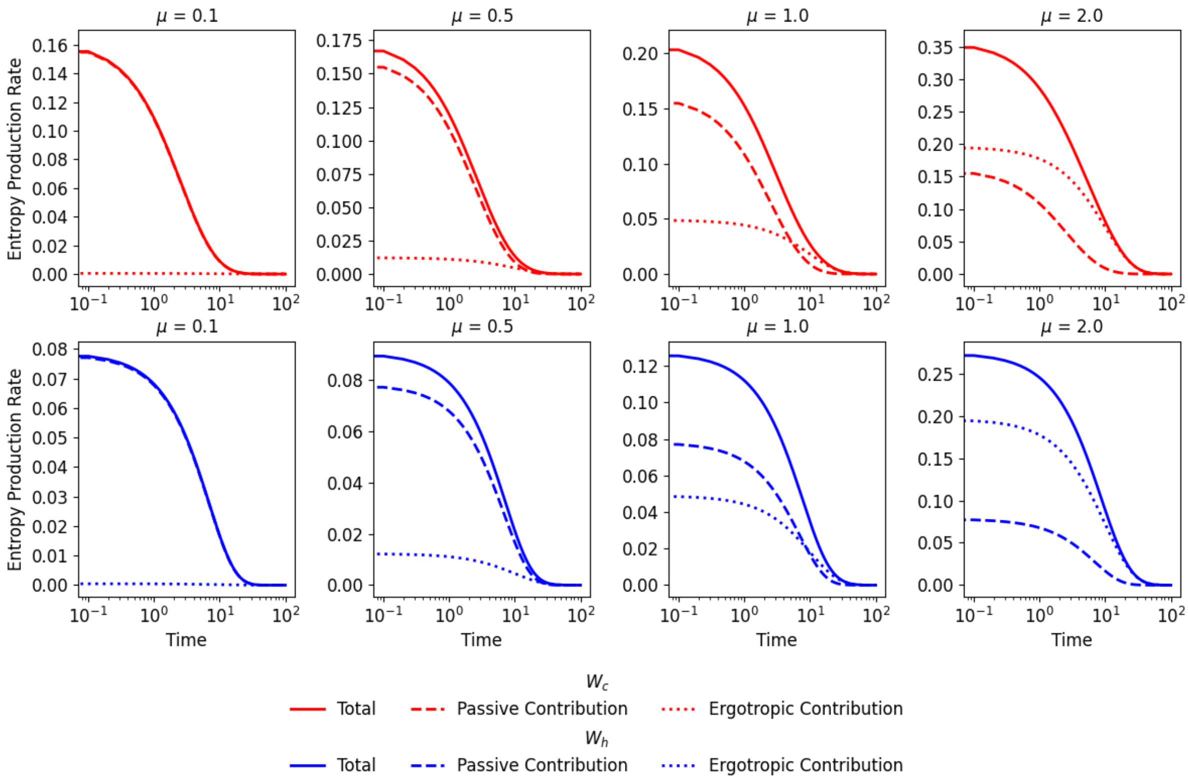

Figure 14 – Total entropy production rate(solid) with its ergotropic (dotted) and passive (dashed) contributions for heating (red) and cooling (blue) equidistant displaced thermal states.

a consequence, in order to make a fair comparison, we must start with two equidistant squeezed states. The ergotropy of a squeezed thermal state is

$$\mathcal{E}(t) = \omega f(\beta_t)(\cosh(2r_t) - 1). \tag{4.63}$$

Because of the dependence of $f(\beta_t)$, applying the same squeezing (that is, the same r_t and ϕ_t) in different thermal states results in a higher ergotropy to the squeezed hotter state. Therefore, to make an even fairer comparison, we will impose that the initial states must have the same initial ergotropy.

The Wigner-Fisher Information of a squeezed thermal state is

$$I_W(t) = 2\dot{S}_W^2 + 8\dot{r}_t^2 + 2\dot{\phi}_t^2 \sinh^2(2r_t). \tag{4.64}$$

Again, obtaining analytical thermal kinematic expressions from Eq. (4.64) will not bring additional information, so we will solve it numerically. From the degree of completion (Fig. 15), the thermal relaxation of W_c is still faster than that of W_h, like in Secs. (4.2.1.1) and (4.2.1.2). However, increasing the squeezing parameter reduces the asymmetry.

To understand why the asymmetry is reduced for the case with a bigger squeezing parameter, we can analyse the entropy production rate. Fig. 16 corresponds to a smaller r and Fig. 17 corresponds to a bigger r. Each graph shows the total entropy production rate (solid line) with its passive (dashed line) and ergotropic (dotted line) contributions. Unlike the displacement operation, the squeezing change affects both passive and ergotropic contributions.

4.2.1.4 Heating a displaced state and cooling a squeezed state

After comparing the thermal relaxation of states with the same quantum resource, this section studies the behaviour of the asymmetry when applying different quantum

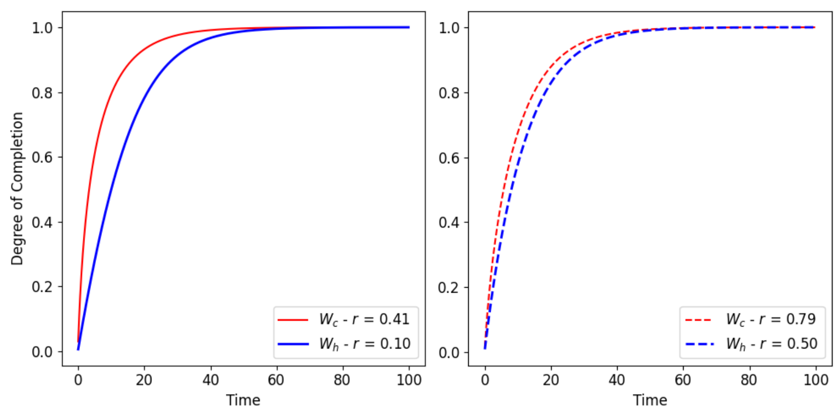

Figure 15 – Degree of completion for a squeezed thermal state, with the right graph using a bigger squeezing parameter than the left graph.

resources to each initial state. Let's apply the displacement on the colder state and the squeezing on the hotter state. The initial states must have the same initial Wigner relative entropy, that is, $K[W_d||W_f] = K[W_s||W_f]$, and the same initial ergotropy, $\mathcal{E}_d = \mathcal{E}_s$.

The degree of completion for this system is presented in Fig. 18, and shows that it is possible to invert the asymmetry depending on the initial state condition. In the left graph, the displacement and squeezing parameters are smaller than in the right graph, and the asymmetry inversion is partial because cooling is faster only in some period of time. In the right graph, where the parameters are bigger, cooling is faster than heating for all times.

To study the entropy production rate of heating, we used the same expressions of Section 4.2.1.2, and for the entropy production rate of cooling, we used the expressions of Section 4.2.1.3. Fig. 19 and Fig. 20 show the ergotropic and passive contributions for the entropy production of smaller and bigger parameters, respectively. The comparison of these graphs indicates that cooling becomes completely faster than heating only when its entropy production is initially higher.

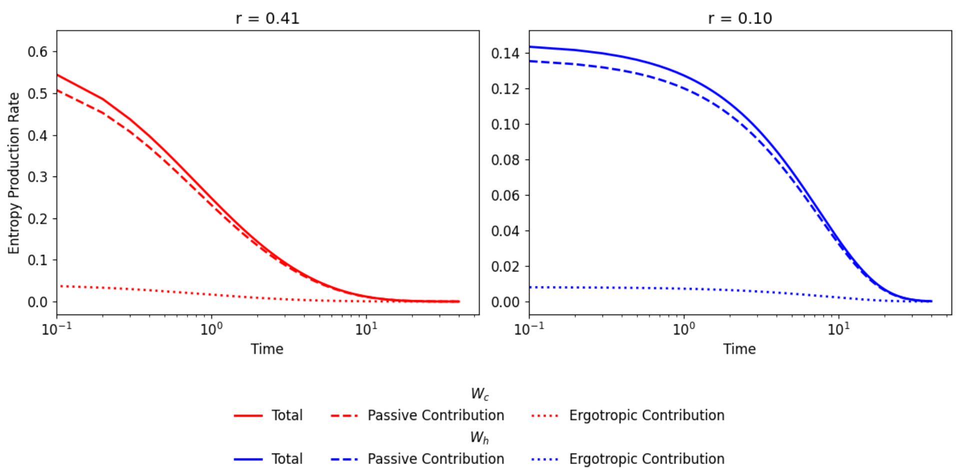

Figure 16 – Entropy production rate for heating (red) and cooling (blue) weakly squeezed thermal state. There are the passive contribution (dashed) and ergotropic contribution (dotted) of the total entropy production rate (solid).

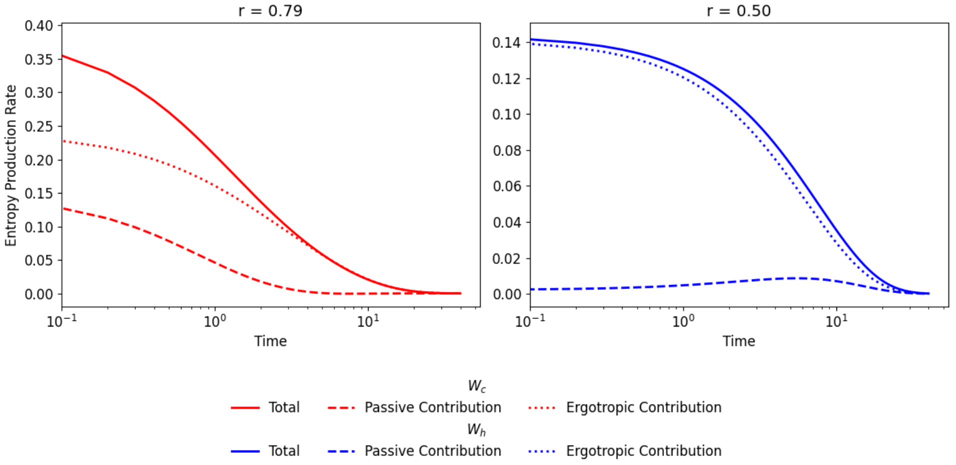

Figure 17 – Entropy production rate for heating (red) and cooling (blue) strongly squeezed thermal states. There are the passive contribution (dashed) and ergotropic contribution (dotted) of the total entropy production rate (solid).

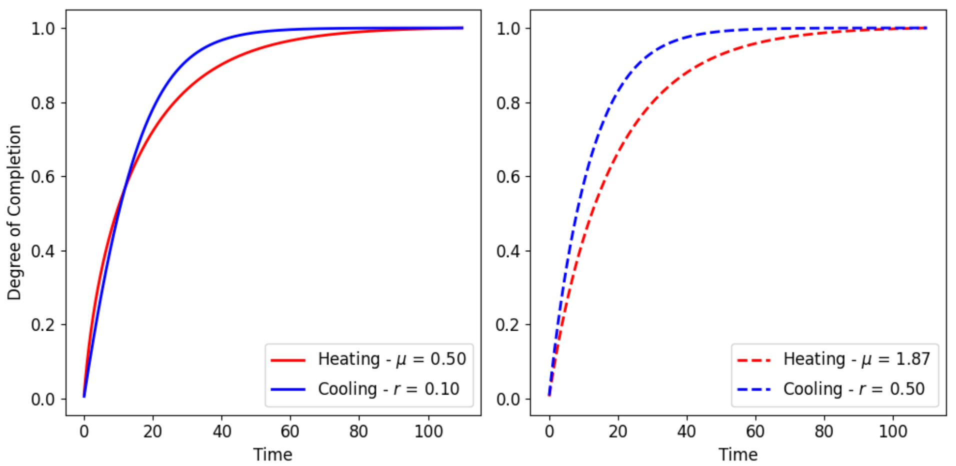

Figure 18 – Degree of completion for heating (red) a displaced thermal state and cooling (blue) a squeezed thermal state.

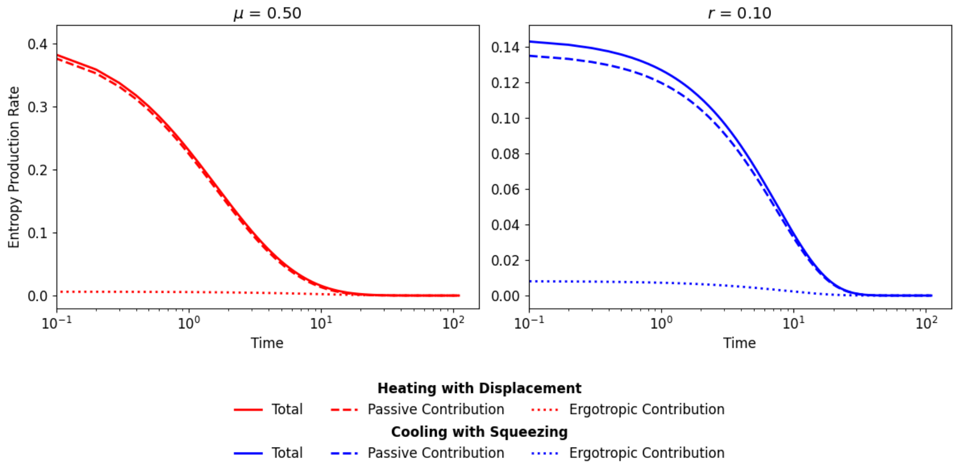

Figure 19 – Entropy production rate for heating (red) a state with small μ and cooling (blue) a state with small r. The passive contribution is in dashed lines, the ergotropic contribution is in dotted lines, and the total entropy production rate is in solid lines.

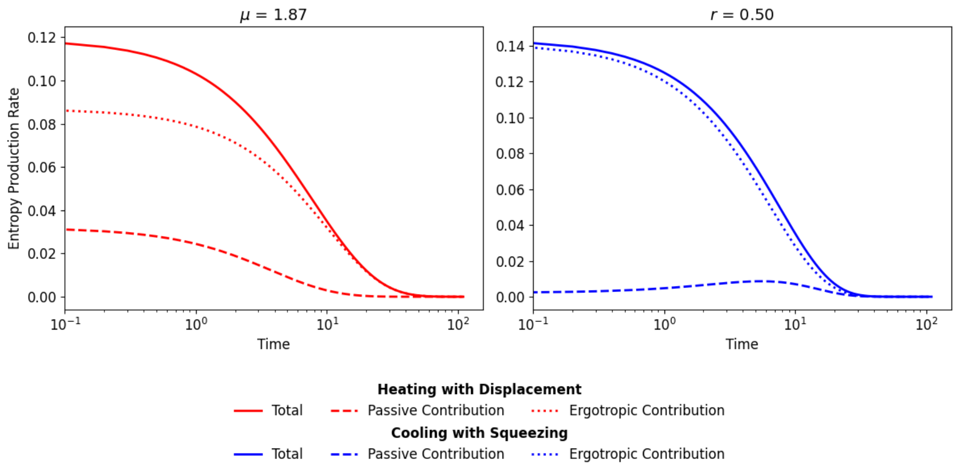

Figure 20 – Entropy production rate for heating (red) a state with big μ and cooling (blue) a state with big r. The passive contribution is in dashed lines, the ergotropic contribution is in dotted lines, and the total entropy production rate is in solid lines.

5 CONCLUSÃO

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APPENDIX

APPENDIX A – APÊNDICE(S)

Elemento opcional, que consiste em texto ou documento elaborado pelo autor, a fim de complementar sua argumentação, conforme a ABNT NBR 14724 (?).

Os apêndices devem ser identificados por letras maiúsculas consecutivas, seguidas de hífen e pelos respectivos títulos. Excepcionalmente, utilizam-se letras maiúsculas dobradas na identificação dos apêndices, quando esgotadas as 26 letras do alfabeto. A paginação deve ser contínua, dando seguimento ao texto principal. (?)

APPENDIX B – EXEMPLO DE TABELA CENTRALIZADA VERTICALMENTE E HORIZONTALMENTE

A Table 2 exemplifica como proceder para obter uma tabela centralizada verticalmente e horizontalmente.

Table 2 – Exemplo de tabela centralizada verticalmente e horizontalmente

Coluna A	**Coluna B**
Coluna A, Linha 1	Este é um texto bem maior para exemplificar como é centralizado verticalmente e horizontalmente na tabela. Segundo parágrafo para verificar como fica na tabela
Quando o texto da coluna A, linha 2 é bem maior do que o das demais colunas	Coluna B, linha 2

Fonte: Elaborada pelos autores.

APPENDIX C – EXEMPLO DE TABELA COM GRADE

A Table 3 exemplifica a inclusão de traços estruturadores de conteúdo para melhor compreensão do conteúdo da tabela, em conformidade com as normas de apresentação tabular do IBGE.

Table 3 – Exemplo de tabelas com grade

Coluna A	**Coluna B**
A1	B1
A2	B2
A3	B3
A4	B4

Fonte: Elaborada pelos autores.

ANNEX

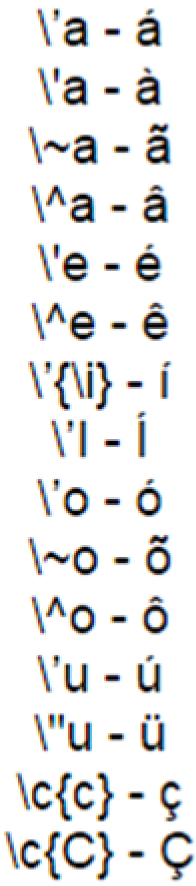
ANNEX A – EXEMPLO DE ANEXO

Elemento opcional, que consiste em um texto ou documento não elaborado pelo autor, que serve de fundamentação, comprovação e ilustração, conforme a ABNT NBR 14724. (**?**).

O **ANEXO B** exemplifica como incluir um anexo em pdf.

ANNEX B – ACENTUAÇÃO (MODO TEXTO - LaTeX)

Figure 21 – Acentuação (modo texto - LaTeX)



Fonte: Comandos [. . .] (??)